

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 1

计算矩阵 $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ 的行列式。

解:

Given $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

按定义或行化简计算行列式:

$$\det(A) = 5$$

最终答案:

5

X 已验证: cofactor_expansion

题 2

求矩阵 $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ 的逆矩阵。

解:

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$\det(A) = -2 \neq 0$, so A is invertible.

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

最终答案:

$$\begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

X 已验证: multiply_identity

题 3

计算矩阵 $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{pmatrix}$ 的行列式。

解:

$$\text{Given } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

按定义或行化简计算行列式:

$$\det(A) = 30$$

最终答案:

$$\boxed{30}$$

X 已验证: cofactor_expansion

题 4

求矩阵 $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ 的逆矩阵。

解:

$$\text{Given } A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$\det(A) = 1 \neq 0$, so A is invertible.

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

最终答案:

$$\boxed{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}$$

X 已验证: multiply_identity

题 5

设 $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$, 计算矩阵乘积 AB 。

解:

$$\text{Given } A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\text{Row-by-column multiplication: } AB = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

最终答案:

$$\begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

X 已验证: dimension_check

题 6

求矩阵 $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ 的秩。

解:

$$\text{Given } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

初等行变换化行阶梯形:

$$\text{rank}(A) = 2$$

最终答案:

$$2$$

X 已验证: rank_nullity

题 7

求矩阵 $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ 的特征值。

解:

$$\text{Given } A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

特征方程: $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \lambda^2 - 7\lambda + 10$$

Solving: $\lambda \in [5, 2]$

Multiplicities: $\lambda=5$ (重数 1), $\lambda=2$ (重数 1)

最终答案:

$$[5, 2]$$

X 已验证: `det_charpoly`

题 8

求矩阵 $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ 的特征值。

解:

$$\text{Given } A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

特征方程: $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \lambda^2 - 6\lambda + 9$$

Solving: $\lambda \in [3]$

Multiplicities: $\lambda=3$ (重数 2)

最终答案:

$$[3, 3]$$

X 已验证: `det_charpoly`

题 9

求矩阵 $A = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$ 的特征多项式与特征值。

解:

$$\text{Given } A = \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix}$$

特征方程: $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \lambda^2 - 10\lambda + 21$$

Solving: $\lambda \in [7, 3]$

Multiplicities: $\lambda=7$ (重数 1), $\lambda=3$ (重数 1)

最终答案:

$$\boxed{[7, 3]}$$

X 已验证: `det_charpoly`

题 10

解线性方程组 $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} x = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$ 。

解:

Given $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$

Gaussian elimination: $x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

最终答案:

$$\boxed{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}$$

X 已验证: `residual_check`

求解汇总

题号	求解器	求解	验证
1	sympy_linalg	X	X
2	sympy_linalg	X	X
3	sympy_linalg	X	X
4	sympy_linalg	X	X
5	sympy_linalg	X	X
6	sympy_linalg	X	X
7	sympy_linalg	X	X
8	sympy_linalg	X	X
9	sympy_linalg	X	X
10	sympy_linalg	X	X

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 1

解微分方程 $y' + 2y = e^{-x}$ 。

解:

Equation: $2y(x) + \frac{d}{dx}y(x) = e^{-x}$

识别为 1 阶常微分方程，调用 SymPy dsolve 符号求解。

General solution: $y(x) = (C_1 e^{-x} + 1) e^{-x}$

非齐次项为 $g(x) \neq 0$: 通解 = 齐次通解 + 特解。

最终答案:

$$(C_1 e^{-x} + 1) e^{-x}$$

X 已验证: ode_residual

题 2

解微分方程 $y'' - 5y' + 6y = 0$ 。

解:

Equation: $6y(x) - 5\frac{d}{dx}y(x) + \frac{d^2}{dx^2}y(x) = 0$

识别为 2 阶常微分方程，调用 SymPy dsolve 符号求解。

General solution: $y(x) = (C_1 + C_2 e^x) e^{2x}$

最终答案:

$$(C_1 + C_2 e^x) e^{2x}$$

X 已验证: ode_residual

题 3

解微分方程 $y'' + y = 0$ 。

解:

Equation: $y(x) + \frac{d^2}{dx^2}y(x) = 0$

识别为 2 阶常微分方程，调用 SymPy dsolve 符号求解。

General solution: $y(x) = C_1 \sin(x) + C_2 \cos(x)$

最终答案:

$$C_1 \sin(x) + C_2 \cos(x)$$

X 已验证: ode_residual

题 4

解微分方程 $y' = xy$ 。

解:

Equation: $\frac{d}{dx}y(x) = xy(x)$

识别为 1 阶常微分方程, 调用 SymPy dsolve 符号求解。

General solution: $y(x) = C_1 e^{\frac{x^2}{2}}$

非齐次项为 $g(x) \neq 0$: 通解 = 齐次通解 + 特解。

最终答案:

$$C_1 e^{\frac{x^2}{2}}$$

X 已验证: ode_residual

题 5

解微分方程 $y'' - y = e^x$ 。

解:

Equation: $-y(x) + \frac{d^2}{dx^2}y(x) = e^x$

识别为 2 阶常微分方程, 调用 SymPy dsolve 符号求解。

General solution: $y(x) = C_2 e^{-x} + \left(C_1 + \frac{x}{2}\right) e^x$

非齐次项为 $g(x) \neq 0$: 通解 = 齐次通解 + 特解。

最终答案:

$$C_2 e^{-x} + \left(C_1 + \frac{x}{2}\right) e^x$$

X 已验证: ode_residual

题 6

解微分方程 $\frac{dy}{dx} - 3y = 0$ 。

解:

Equation: $-3y(x) + \frac{d}{dx}y(x) = 0$

识别为 1 阶常微分方程，调用 SymPy dsolve 符号求解。

General solution: $y(x) = C_1e^{3x}$

最终答案:

$$C_1e^{3x}$$

X 已验证: ode_residual

求解汇总

题号	求解器	求解	验证
1	sympy_ode	X	X
2	sympy_ode	X	X
3	sympy_ode	X	X
4	sympy_ode	X	X
5	sympy_ode	X	X
6	sympy_ode	X	X

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 PartA1

Find the derivative of $x^3 + 2x^2 - 5x + 1$.

解:

Let $f(x) = x^3 + 2x^2 - 5x + 1$

Apply differentiation rules: $f'(x) = 3x^2 + 4x - 5$

最终答案:

$$3x^2 + 4x - 5$$

X 已验证: derivative_equivalence

题 PartA2

Evaluate the integral $\int x^2 \sin(x) dx$.

解:

$$\int x^2 \sin(x) dx = -x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C$$

最终答案:

$$-x^2 \cos(x) + 2x \sin(x) + 2 \cos(x)$$

X 已验证: integrand_recovery

题 PartA3

Compute the limit $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.

解:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

最终答案:

$$1$$

X 已验证: numeric_limit

题 PartA4

Find the derivative of $e^{x^2} \cdot \ln(x)$.

解:

Let $f(x) = e^{x^2} \log(x)$

Apply differentiation rules: $f'(x) = 2xe^{x^2} \log(x) + \frac{e^{x^2}}{x}$

最终答案:

$$2xe^{x^2} \log(x) + \frac{e^{x^2}}{x}$$

X 已验证: derivative_equivalence

题 PartA5

Evaluate $\int_0^1 \frac{1}{1+x^2} dx$.

解:

$$\int_0^1 \frac{1}{x^2 + 1} dx = \frac{\pi}{4}$$

最终答案:

$$\frac{\pi}{4}$$

X 已验证: numeric_quad

题 PartB6

Solve the equation $x^2 - 5x + 6 = 0$.

解:

Equation: $x^2 - 5x + 6 = 0$

Solving for x : roots = $[2, 3]$

最终答案:

$$[2, 3]$$

X 已验证: substitute_roots

题 PartB7

Solve $x^3 - 6x^2 + 11x - 6 = 0$.

解:

Equation: $x^3 - 6x^2 + 11x - 6 = 0$

Solving for x : roots = [1, 2, 3]

最终答案:

[1, 2, 3]

X 已验证: substitute_roots

题 PartC8

Compute the determinant of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

解:

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

按定义或行化简计算行列式:

$$\det(A) = -2$$

最终答案:

-2

X 已验证: cofactor_expansion

题 PartC9

Find the eigenvalues of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

解:

Given $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

特征方程: $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = \lambda^2 - 4\lambda + 3$$

Solving: $\lambda \in [3, 1]$

Multiplicities: $\lambda=3$ (重数 1), $\lambda=1$ (重数 1)

最终答案:

$$[3, 1]$$

X 已验证: det_charpoly

题 PartD10

Solve the ODE $y' + 2y = e^{-x}$.

解:

Equation: $2y(x) + \frac{d}{dx}y(x) = e^{-x}$

识别为 1 阶常微分方程, 调用 SymPy dsolve 符号求解。

General solution: $y(x) = (C_1 e^{-x} + 1) e^{-x}$

非齐次项为 $g(x) \neq 0$: 通解 = 齐次通解 + 特解。

最终答案:

$$(C_1 e^{-x} + 1) e^{-x}$$

X 已验证: ode_residual

求解汇总

题号	求解器	求解	验证
PartA1	sympy_derivative	X	X
PartA2	sympy_integral	X	X
PartA3	sympy_limit	X	X
PartA4	sympy_derivative	X	X
PartA5	sympy_integral	X	X
PartB6	sympy_equation	X	X
PartB7	sympy_equation	X	X
PartC8	sympy_linalg	X	X
PartC9	sympy_linalg	X	X
PartD10	sympy_ode	X	X

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 Q1

Can the graph of a function have more than one tangent at a given point? If so, graph your answer. If not, explain why.

解:

In general, a smooth function has exactly **one** tangent line at each point where the derivative exists.

However, a *curve* (not necessarily the graph of a function) can have **multiple tangents at a self-intersection point** —e.g. a figure-eight curve.

For the graph of a single function f : the tangent at a is unique whenever $f'(a)$ exists; if $f'(a)$ does not exist there may be none (corner) or a vertical tangent.

最终答案:

A function graph has at most one tangent where $f'(a)$ exists; curves can have several at self-intersections.

题 Q2

Is there a function whose graph doesn't have a tangent at some point? If so, graph your answer. If not, explain why.

解:

Yes —a function fails to have a tangent where it is not differentiable.

Examples: $f(x) = |x|$ at $x = 0$ (corner: left slope -1 , right slope $+1$); $f(x) = x^{1/3}$ at $x = 0$ (vertical tangent —slope unbounded).

At such points the tangent-line limit $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$ does not exist.

最终答案:

Yes: $f(x)=|x|$ at $x=0$ (corner) has no tangent.

题 P1

Suppose δ is a positive real number. How do you describe all real numbers x that are within δ of 0?

解:

x within δ of 0: $|x| < \delta$, i.e. $-\delta < x < \delta$

(a) Using inequalities ($<$, $>$).

Using inequalities:

$$-\delta < x < \delta$$

$$\boxed{-\delta < x < \delta}$$

(b) Using absolute value notation.

$$|x| < \delta$$

$$\boxed{|x| < \delta}$$

(c) Using interval notation.

$$(-\delta, \delta)$$

$$\boxed{(-\delta, \delta)}$$

最终答案: $|x| < \delta$, i.e. $-\delta < x < \delta$

题 P2

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a ?

解:

x within δ of 0: $|x| < \delta$, i.e. $-\delta < x < \delta$

(a) Graphically, as on the line above.

[数轴示意: 中心与端点均为开区间]

See graph

(b) Using inequalities. There are at least two ways to do this. Find both.

Using inequalities:

$$-\delta < x < \delta$$

$$\boxed{-\delta < x < \delta}$$

(c) Using absolute value notation.

$$|x| < \delta$$

$$|x| < \delta$$

(d) Using interval notation.

$$(-\delta, \delta)$$

$$(-\delta, \delta)$$

最终答案: $|x| < \delta$, i.e. $-\delta < x < \delta$

题 P3

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a , but not equal to a :

解:

x within δ of 0, not equal to 0: $0 < |x| < \delta$

(a) Graphically.

[数轴示意: 中心与端点均为开区间]

See graph

(b) Using inequalities.

Using inequalities: $x \neq 0$

$$-\delta < x < 0 \text{ or } 0 < x < \delta$$

$$-\delta < x < 0 \text{ or } 0 < x < \delta$$

(c) Using absolute value notation.

$$0 < |x| < \delta$$

$$0 < |x| < \delta$$

(d) Using interval notation.

$$(-\delta, 0) \cup (0, \delta)$$

$$(-\delta, 0) \cup (0, \delta)$$

最终答案:

$$0 < |x| < \delta$$

题 P4

Let $f(x) = x^2$.

解:

Given: $f(x) = x^2$

(a) Find all the positive numbers x such that $f(x)$ is within 1 of 9.

Need: $9.0 - 1 < x^2 < 9.0 + 1$

i.e. $8.0 < x^2 < 10.0$

Solving: $x \in (2.82842712474619, 3.16227766016838)$ (positive x only)

$(2.82842712474619, 3.16227766016838)$

(b) Find a real number δ such that whenever x is within δ of 3, $f(x)$ is within 1 of 9.

Need: $|x - 3| < \delta \Rightarrow |x^2 - 9.0| < 1$

Boundaries: $f(x) = 9.0 \pm 1$ gives $x \in [-3.16227766016838, 3.16227766016838, -2.82842712474619]$

Minimum distance from $x = 3$: $\delta = 0.16227766016838$

$\delta = 0.16227766016838$

(c) Find a real number δ such that whenever x is within δ of 3, $f(x)$ is within $1/2$ of 9.

(无法解析 - 计算子题)

(d) Is it true that for any positive number ε , there is a positive δ so that $f(x)$ is within ...

Yes — f is continuous at a , so for any $\varepsilon > 0$ there exists $\delta > 0$ with $|x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Yes (continuity)

最终答案:

$$f(x) = x^2$$

题 AP1

Graph $y = 2x - x^2$. For which points a is the tangent line to the curve at the point $(a, 2a - a^2)$ a horizontal line? How is the value of $2x - x^2$ at these points related to the values of $2x - x^2$ at other points?

解:

Given: $y = -x^2 + 2x$

Derivative: $y' = 2 - 2x$

Horizontal tangent where $y' = 0$: $2 - 2x = 0$

Point: $((1, 1))$

最终答案:

$$(1, 1)$$

题 AP2

The point $(2, 0)$ is on the curve $y = 2x - x^2$.

解:

Given: $y = -x^2 + 2x$

Derivative: $y' = 2 - 2x$

At point $(2, 0)$:

Slope: $f'(2) = -2$

Tangent: $y = 4 - 2x$

(a) Draw the tangent line to the curve at the point $(2, 0)$.

画图题: 用图形工具绘出切线。

See graph

(b) Try to determine the equation of this tangent line.

(无法识别子题类型)

最终答案:

$$y = 4 - 2x$$

求解汇总

题号	求解器	求解	验证
Q1	conceptual_template	X	—
Q2	conceptual_template	X	—
P1	epsilon_delta_template	X	—
P2	epsilon_delta_template	X	—
P3	epsilon_delta_template	X	—
P4	epsilon_delta_computation	X	—
AP1	sympy_tangent	X	—
AP2	sympy_tangent	X	—

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 Q1

Suppose f is a function and L and a are real numbers.

解:

See sub-problems.

(a) Describe what is meant by $\lim_{x \rightarrow a} f(x) = L$ as best you can.

(包含抽象函数, 需要 LLM)

(b) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a) = L$.

画图题: 请用图形工具。

See graph

(c) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, but $f(a) \neq L$.

画图题: 请用图形工具。

See graph

(d) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a)$ does not exist.

画图题: 请用图形工具。

See graph

题 Q2

Is ∞ a number? What does $\lim_{x \rightarrow a} f(x) = \infty$ mean? Does $\lim_{x \rightarrow a} f(x) = \infty$ mean that the limit exists?

解:

∞ is **not** a real number.

$\lim_{x \rightarrow a} f(x) = \infty$ means $f(x)$ grows without bound as $x \rightarrow a$ —for every $M > 0$ there is a $\delta > 0$ with $f(x) > M$ whenever $0 < |x - a| < \delta$.

The limit does **not exist** in the usual sense; writing “ $= \infty$ ” describes the *manner of divergence*, not a limit value.

最终答案:

∞ is not a number. $\lim = \infty$ means unbounded growth (DNE), not an existing limit.

题 Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

解:

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a), and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Intuition: f is squeezed between two functions converging to L .

Application: $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$ since $-x^2 \leq x^2 \sin(1/x) \leq x^2$.

最终答案:

If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$.

题 Q4

What do we mean by $\lim_{x \rightarrow a^-} f(x) = L$? What is meant by $\lim_{x \rightarrow a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Find an algebraic expression for such a function.

解:

$\lim_{x \rightarrow a^-} f(x) = L$: **left-hand limit** —approach a from below (through values less than a).

$\lim_{x \rightarrow a^+} f(x) = L$: **right-hand limit** —approach a from above.

Two-sided limit exists $\iff$ both one-sided limits exist and are equal.

Example: $f(x) = \begin{cases} x & x < 0 \\ x + 1 & x \geq 0 \end{cases}$ at $a = 0$: $\lim_{x \rightarrow 0^-} = 0 \neq 1 = \lim_{x \rightarrow 0^+}$, so the limit

DNE.

最终答案:

Left: approach from below; right: from above; two-sided exists iff both agree.

题 P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exist.

解:

No —the **Sum Law** requires both limits to exist.

Counterexample: $f(x) = \frac{1}{x}$, $g(x) = -\frac{1}{x}$ at $x = 0$: neither limit exists, yet $\lim[f+g] = \lim 0 = 0$.

最终答案:

No. $f=1/x$, $g=-1/x$: sum limit = 0 but individual limits DNE.

题 P2

(a) Find $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x \rightarrow 0^-} (\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{|x|})$, if they exist.

解:

$$\lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Left: } \lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Right: } \lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Since $-\infty \neq 0$, the two-sided limit **does not exist**.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the two-sided limit **does not exist**.

(b) Show that $\lim_{x \rightarrow 0} |x| = 0$.

(无法解析子题极限)

(c) Show that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the limit DNE.

DNE

最终答案:

0, $-\infty$, DNE, DNE

题 P3

For what kinds of functions can one evaluate the limit $\lim_{x \rightarrow a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \rightarrow a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

解:

Continuous functions allow direct substitution: $\lim_{x \rightarrow a} f(x) = f(a)$.

Includes polynomials, rational functions (where denominator $\neq 0$), $\sin x$, $\cos x$, e^x , $\ln x$ on their domains.

Counterexample needing algebra first: $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ gives $0/0$ by substitution but $= 2$ after cancellation.

最终答案:

Continuous functions allow direct substitution; $0/0$ forms need algebra first.

题 AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

解:

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

(a) $\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24}$

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\boxed{-20}$$

(b) $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4}$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\boxed{10}$$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

$$\boxed{0}$$

最终答案:

$$\boxed{-20, 10, 0}$$

X 已验证: numeric_limit

题 AP2

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

解:

$$\lim_{x \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$$

最终答案:

$$\boxed{\pi}$$

X 已验证: numeric_limit

题 AP3

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

未自动求解: 无法提取极限表达式

求解汇总

题号	求解器	求解	验证
Q1	sympy_limit	X	—
Q2	conceptual_template	X	—
Q3	conceptual_template	X	—
Q4	conceptual_template	X	—
P1	conceptual_template	X	—
P2	sympy_limit	X	—
P3	conceptual_template	X	—
AP1	sympy_limit	X	X
AP2	sympy_limit	X	X
AP3	none	×	—

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 Q1

Suppose f is a function and L and a are real numbers.

解:

See sub-problems.

(a) Describe what is meant by $\lim_{x \rightarrow a} f(x) = L$ as best you can.

(包含抽象函数, 需要 LLM)

(b) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a) = L$.

画图题: 请用图形工具。

See graph

(c) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, but $f(a) \neq L$.

画图题: 请用图形工具。

See graph

(d) Graph a function $y = f(x)$ such that $\lim_{x \rightarrow a} f(x) = L$, and $f(a)$ does not exist.

画图题: 请用图形工具。

See graph

题 Q2

Is ∞ a number? What does $\lim_{x \rightarrow a} f(x) = \infty$ mean? Does $\lim_{x \rightarrow a} f(x) = \infty$ mean that the limit exists?

解:

∞ is **not** a real number.

$\lim_{x \rightarrow a} f(x) = \infty$ means $f(x)$ grows without bound as $x \rightarrow a$ —for every $M > 0$ there is a $\delta > 0$ with $f(x) > M$ whenever $0 < |x - a| < \delta$.

The limit does **not exist** in the usual sense; writing “ $= \infty$ ” describes the *manner of divergence*, not a limit value.

最终答案:

∞ is not a number. $\lim = \infty$ means unbounded growth (DNE), not an existing limit.

题 Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

解:

Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a), and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Intuition: f is squeezed between two functions converging to L .

Application: $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$ since $-x^2 \leq x^2 \sin(1/x) \leq x^2$.

最终答案:

If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$.

题 Q4

What do we mean by $\lim_{x \rightarrow a^-} f(x) = L$? What is meant by $\lim_{x \rightarrow a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Find an algebraic expression for such a function.

解:

$\lim_{x \rightarrow a^-} f(x) = L$: **left-hand limit** —approach a from below (through values less than a).

$\lim_{x \rightarrow a^+} f(x) = L$: **right-hand limit** —approach a from above.

Two-sided limit exists $\iff$ both one-sided limits exist and are equal.

Example: $f(x) = \begin{cases} x & x < 0 \\ x + 1 & x \geq 0 \end{cases}$ at $a = 0$: $\lim_{x \rightarrow 0^-} = 0 \neq 1 = \lim_{x \rightarrow 0^+}$, so the limit

DNE.

最终答案:

Left: approach from below; right: from above; two-sided exists iff both agree.

题 P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exist.

解:

No —the **Sum Law** requires both limits to exist.

Counterexample: $f(x) = \frac{1}{x}$, $g(x) = -\frac{1}{x}$ at $x = 0$: neither limit exists, yet $\lim[f+g] = \lim 0 = 0$.

最终答案:

No. $f=1/x$, $g=-1/x$: sum limit = 0 but individual limits DNE.

题 P2

(a) Find $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x \rightarrow 0^-} (\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{|x|})$, if they exist.

解:

$$\lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Left: } \lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\text{Right: } \lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Since $-\infty \neq 0$, the two-sided limit **does not exist**.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the two-sided limit **does not exist**.

(b) Show that $\lim_{x \rightarrow 0} |x| = 0$.

(无法解析子题极限)

(c) Show that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

$$\text{Left: } \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

$$\text{Right: } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since $-1 \neq 1$, the limit DNE.

DNE

最终答案:

0, $-\infty$, DNE, DNE

题 P3

For what kinds of functions can one evaluate the limit $\lim_{x \rightarrow a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \rightarrow a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

解:

Continuous functions allow direct substitution: $\lim_{x \rightarrow a} f(x) = f(a)$.

Includes polynomials, rational functions (where denominator $\neq 0$), $\sin x$, $\cos x$, e^x , $\ln x$ on their domains.

Counterexample needing algebra first: $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ gives $0/0$ by substitution but $= 2$ after cancellation.

最终答案:

Continuous functions allow direct substitution; $0/0$ forms need algebra first.

题 AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

解:

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

(a) $\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24}$

$$\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7}x^5 - 24} = -20$$

$$\boxed{-20}$$

(b) $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4}$

$$\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$$

$$\boxed{10}$$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

$$\boxed{0}$$

最终答案:

$$\boxed{-20, 10, 0}$$

X 已验证: numeric_limit

题 AP2

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

解:

$$\lim_{x \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$$

最终答案:

$$\boxed{\pi}$$

X 已验证: numeric_limit

题 AP3

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

解:

[MOCK] 这是 LLM MOCK=1 模式下的确定性响应，用于流水线集成测试。

[MOCK] 真实运行请配置 QWEN_API_KEY / KIMI_API_KEY / DEEPSEEK_API_KEY。

最终答案:

$\boxed{(\text{mock})}$

△ LLM 推理结果，未经独立数值验证，请人工复核

求解汇总

题号	求解器	求解	验证
Q1	sympy_limit	X	—
Q2	conceptual_template	X	—
Q3	conceptual_template	X	—
Q4	conceptual_template	X	—
P1	conceptual_template	X	—
P2	sympy_limit	X	—
P3	conceptual_template	X	—
AP1	sympy_limit	X	X
AP2	sympy_limit	X	X
AP3	llm	X	—

注: 本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。

作业解答 Homework Solutions

线性代数

谢嘉乐 | 2026-10-05

题 1

求 $\lim_{x \rightarrow 0} \frac{|x|}{x}$ 。

解:

Left: $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$ Right: $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$ Since $-1 \neq 1$, the two-sided limit **does not exist**.

最终答案:

DNE

题 2

求 $\lim_{x \rightarrow 0^+} \frac{1}{x}$ 。

解:

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

最终答案:

 ∞

X 已验证: numeric_limit

题 3

判断 $\lim_{x \rightarrow \infty} \sin(x)$ 是否存在。

解:

 $\lim_{x \rightarrow \infty} \sin(x)$ oscillates within $\langle -1, 1 \rangle$.The function does not settle to a single value, so the limit **does not exist**.

最终答案:

DNE

题 4

计算 $\int \frac{1}{x} dx$ 。

解:

$$\int \frac{1}{x} dx = \log(x) + C$$

最终答案:

$$\log(x)$$

X 已验证: integrand_recovery

题 5

解方程 $\frac{\sqrt{x@bad+f}}{2} = 5$ 。

未自动求解: 无法提取方程

题 6

求矩阵 $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ 的逆矩阵。

未自动求解: $\det(A)=0$, 矩阵不可逆

题 7

解微分方程 $y'' + 4y = 0$ 。

解:

Equation: $4y(x) + \frac{d^2}{dx^2}y(x) = 0$

识别为 2 阶常微分方程, 调用 SymPy dsolve 符号求解。

General solution: $y(x) = C_1 \sin(2x) + C_2 \cos(2x)$

最终答案:

$$C_1 \sin(2x) + C_2 \cos(2x)$$

X 已验证: ode_residual

求解汇总

题号	求解器	求解	验证
1	sympy_limit	X	—
2	sympy_limit	X	X
3	sympy_limit	X	—
4	sympy_integral	X	X
5	none	×	—
6	none	×	—
7	sympy_ode	X	X

注：本文档由 homework-solver 流水线自动生成(确定性符号计算, AI 使用记录见 ai_usage_log.jsonl 与交付文档)。